Unit 5: The Pythagorean Theorem

Practice Problems — All Four Sections

Name: _____ Date: _____ Period: _____

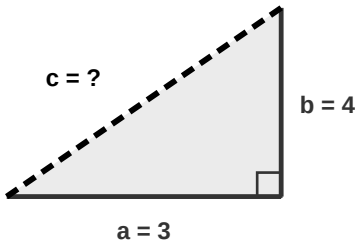
Key Formula: $a^2 + b^2 = c^2$ (a, b = legs; c = hypotenuse — always opposite the 90° angle)

5.1 — Introduction & How to Use

Determine if the missing side is a LEG or the HYPOTENUSE first, then solve.

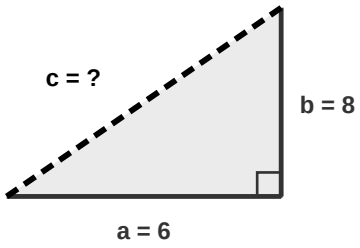
Part A — Solve for the Hypotenuse

Q1. A right triangle has legs 3 and 4. Find the hypotenuse.



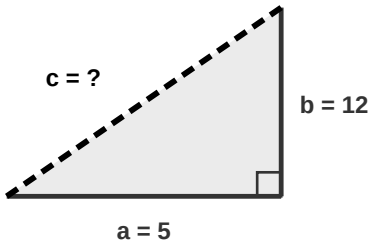
c =: _____

Q2. A right triangle has legs 6 and 8. Find the hypotenuse.



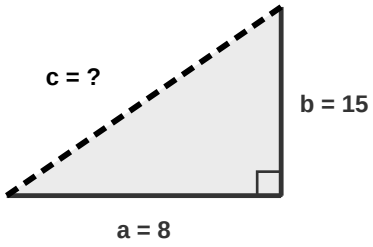
c =: _____

Q3. A right triangle has legs 5 and 12. Find the hypotenuse.



c =: _____

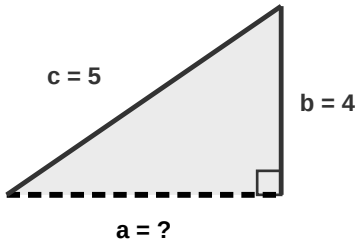
Q4. A right triangle has legs 8 and 15. Find the hypotenuse.



c =: _____

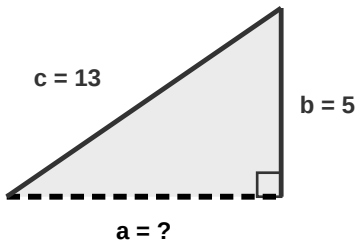
Part B — Solve for a Missing Leg

Q5. Hypotenuse = 5, leg = 4. Find the missing leg.



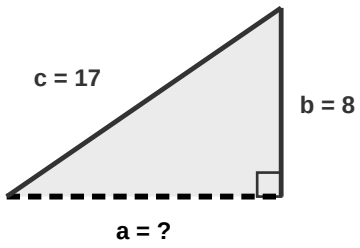
a =: _____

Q6. Hypotenuse = 13, leg = 5. Find the missing leg.



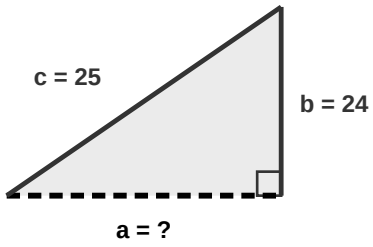
a =: _____

Q7. Hypotenuse = 17, leg = 8. Find the missing leg.



a =: _____

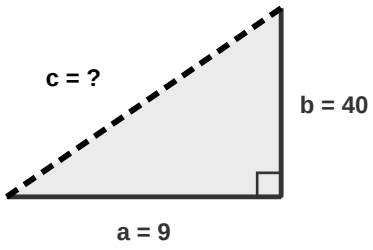
Q8. Hypotenuse = 25, leg = 24. Find the missing leg.



a =: _____

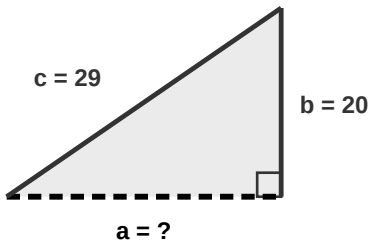
Part C — Identify the Missing Side, Then Solve

Q9. Legs are 9 and 40. Find the hypotenuse.



c =: _____

Q10. Hypotenuse = 29, leg = 20. Find the other leg.



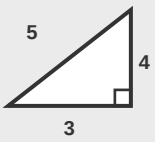
a =: _____

5.2 — Converse of the Pythagorean Theorem

Sort sides smallest to largest. Test: does $a^2 + b^2 = c^2$?

Part A — Classic Triples

Q1. Sides: 3, 4, 5. Is this a right triangle? Show your work.



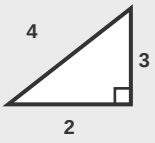
$3^2 + 4^2$ = ___

5^2 = ___

Equal? Yes / No

Yes / No: _____

Q2. Sides: 2, 3, 4. Is this a right triangle? Show your work.



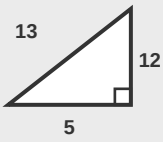
$2^2 + 3^2$ = ___

4^2 = ___

Equal? Yes / No

Yes / No: _____

Q3. Sides: 5, 12, 13. Is this a right triangle? Show your work.



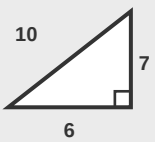
$5^2 + 12^2$ = ___

13^2 = ___

Equal? Yes / No

Yes / No: _____

Q4. Sides: 6, 7, 10. Is this a right triangle? Show your work.



$6^2 + 7^2$ = ___

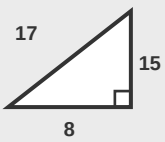
10^2 = ___

Equal? Yes / No

Yes / No: _____

Part B — Larger Numbers

Q5. Sides: 8, 15, 17. Is this a right triangle? Show your work.



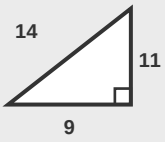
$8^2 + 15^2 =$ ___

$17^2 =$ ___

Equal? Yes / No

Yes / No: _____

Q6. Sides: 9, 11, 14. Is this a right triangle? Show your work.



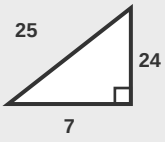
$9^2 + 11^2 =$ ___

$14^2 =$ ___

Equal? Yes / No

Yes / No: _____

Q7. Sides: 7, 24, 25. Is this a right triangle? Show your work.



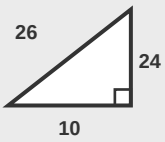
$7^2 + 24^2 =$ ___

$25^2 =$ ___

Equal? Yes / No

Yes / No: _____

Q8. Sides: 10, 24, 26. Is this a right triangle? Show your work.



$10^2 + 24^2 =$ ___

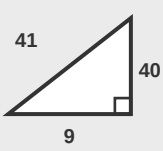
$26^2 =$ ___

Equal? Yes / No

Yes / No: _____

Part C — Tricky Values

Q9. Sides: 9, 40, 41. Is this a right triangle? Show your work.



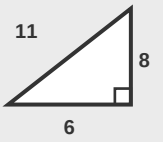
$9^2 + 40^2 =$ ___

$41^2 =$ ___

Equal? Yes / No

Yes / No: _____

Q10. Sides: 6, 8, 11. Is this a right triangle? Show your work. *(Careful — is this the same as 6-8-10?)*



$6^2 + 8^2 =$ ___

$11^2 =$ ___

Equal? Yes / No

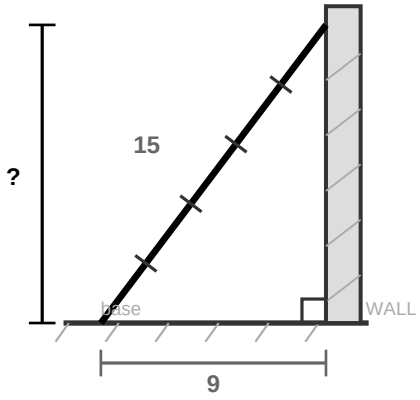
Yes / No: _____

5.3 — Word Problems

Label the diagram, identify the hypotenuse, then set up $a^2 + b^2 = c^2$ and solve.

Part A — Ladders, Ramps & Screens

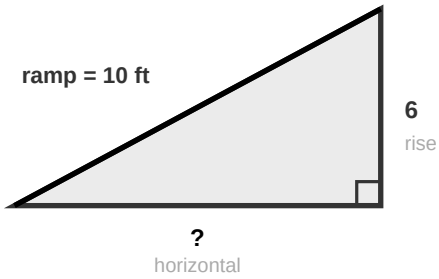
Q1. A ladder 15 ft long leans against a wall. The base is 9 ft from the wall. How high up the wall does it reach?



Height on wall =: _____

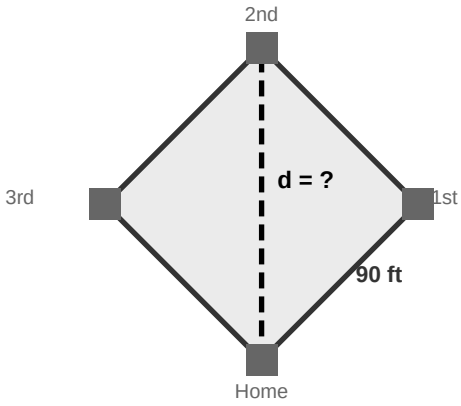
Wheelchair Ramp

Q2. A wheelchair ramp is 10 ft long and rises 6 ft vertically. What is the horizontal length?



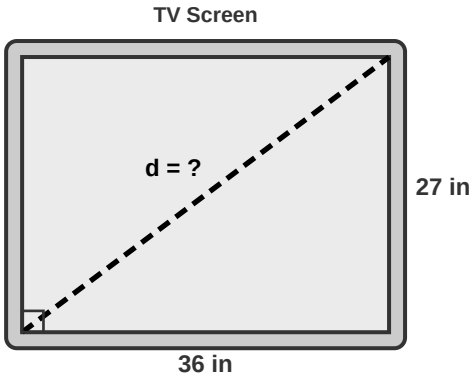
Horizontal length =: _____

Q3. A baseball diamond is a square with 90 ft sides. What is the distance from home plate to 2nd base? Round to the nearest foot.



Distance (nearest ft) =: _____

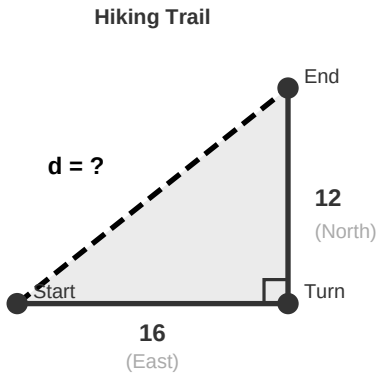
Q4. A TV screen is 36 in wide and 27 in tall. What is the diagonal measurement?



Diagonal =: _____

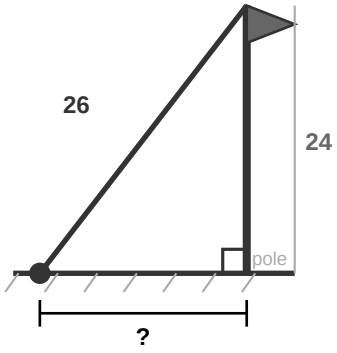
Part B — Navigation & Distance

Q5. A hiker walks 16 km East, then 12 km North. How far is she from the start in a straight line?



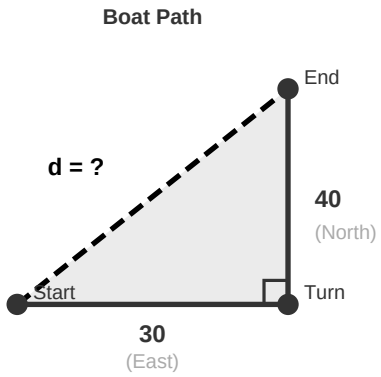
Straight-line distance =: _____

Q6. A flagpole is 24 ft tall. A wire runs from the top down to a stake in the ground. The wire is 26 ft long. How far is the stake from the base of the pole?



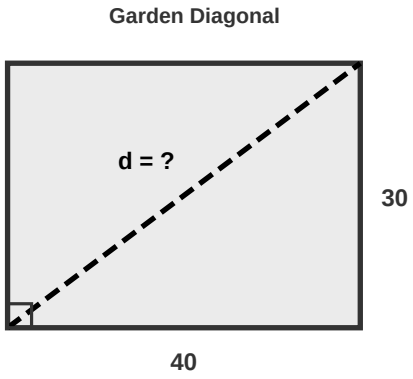
Distance from base =: _____

Q7. A boat travels 30 km East then 40 km North. How far is it from the dock in a straight line?



Straight-line distance =: _____

Q8. A rectangular garden is 40 m wide and 30 m long. How long is the diagonal path from one corner to the opposite corner?



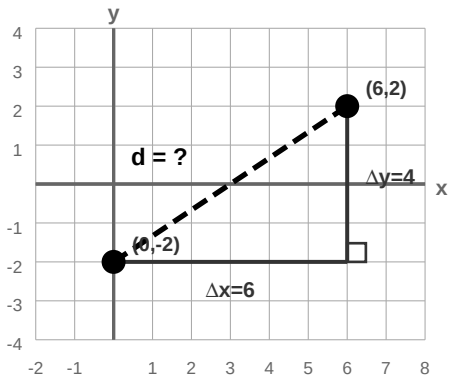
Diagonal =: _____

5.4 — Distance on the Coordinate Plane

Use the two points to form a right triangle, then apply the Pythagorean Theorem to find the distance.

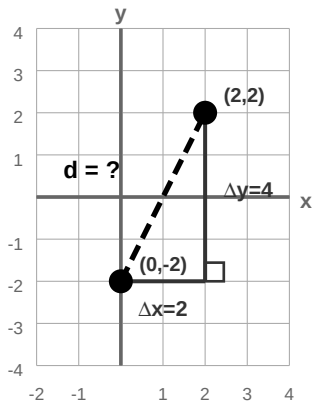
Part A — Integer Coordinates

Q1. Find the distance between (0, -2) and (6, 2). ∆x = ___ ∆y = ___



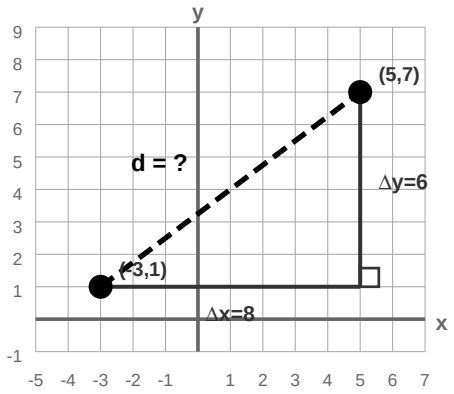
Distance: _____

Q2. Find the distance between (0, -2) and (2, 2). ∆x = ___ ∆y = ___



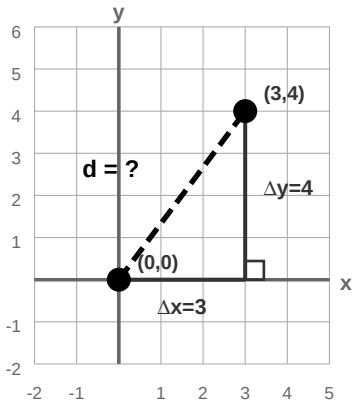
Distance: _____

Q3. Find the distance between (-3, 1) and (5, 7). ∆x = ___ ∆y = ___



Distance: _____

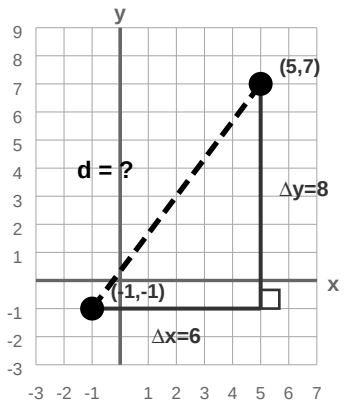
Q4. Find the distance between (0, 0) and (3, 4). ∆x = ___ ∆y = ___



Distance: _____

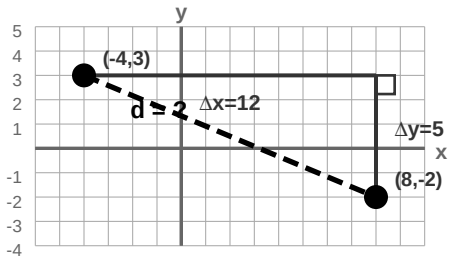
Part B — Negative Coordinates

Q5. Find the distance between (-1, -1) and (5, 7). ∆x = ___ ∆y = ___



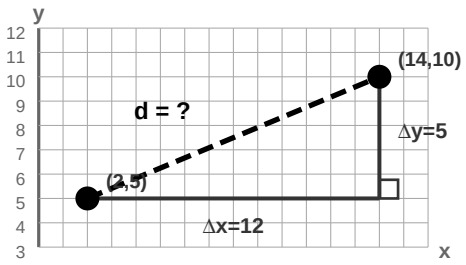
Distance: _____

Q6. Find the distance between (-4, 3) and (8, -2). ∆x = ___ ∆y = ___



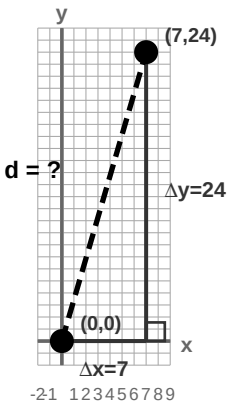
Distance: _____

Q7. Find the distance between (2, 5) and (14, 10). Each unit = 1 mile. ∆x = ___ ∆y = ___



Distance: _____

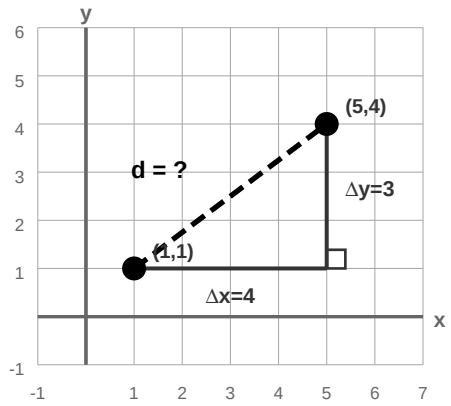
Q8. Find the distance between (0, 0) and (7, 24). ∆x = ___ ∆y = ___



Distance: _____

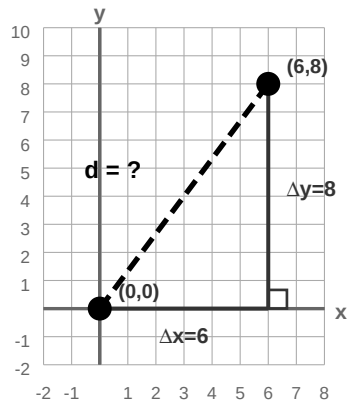
Part C — Applications

Q9. Find the distance between (1, 1) and (5, 4). Triangle A(1,1) B(5,1) C(5,4). Find AC. ∆x = ___ ∆y = ___



Distance: _____

Q10. Find the distance between (0, 0) and (6, 8). P(0,0), Q(6,0), R(6,8). Find PR. ∆x = ___ ∆y = ___



Distance: _____